
Project Proposal: Optimizing Job Shop Scheduling via Group Relative Policy Optimization (GRPO)

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1 Problem Definition

1.1 The Job Shop Scheduling Problem (JSSP)

We consider the standard Job Shop Scheduling Problem, defined by a set of jobs $\mathcal{J} = \{J_1, \dots, J_n\}$ and a set of machines $\mathcal{M} = \{M_1, \dots, M_m\}$. Each job J_i consists of a sequence of operations $O_{i,1} \rightarrow O_{i,2} \dots \rightarrow O_{i,k}$ that must be processed in a strict order (conjunctive constraints). Furthermore, each operation $O_{i,j}$ requires a specific machine and occupies it for a fixed duration p_{ij} , during which no other operation can use that machine (disjunctive constraints).

The objective is to determine a schedule—a start time for every operation—that minimizes the **Makespan** ($C_{\max}$), defined as the completion time of the last operation across all jobs:

$$\text{Minimize } C_{\max} = \max_{i \in \mathcal{J}}(C_i) \quad (1)$$

Following the formulation in [1], we represent the state s_t as a **Disjunctive Graph** $G = (\mathcal{O}, \mathcal{C}, \mathcal{D})$. The scheduling process is treated as a sequential decision-making task where the agent iteratively selects an operation from the set of eligible nodes (those whose predecessors are completed) until all operations are scheduled.

2 Motivation

While JSSP is a well-defined combinatorial problem, solving it efficiently at scale remains an open challenge. Exact solvers (e.g., CP-SAT) are computationally prohibitive for large instances (NP-hard), and heuristic rules (e.g., FIFO, MWKR) often yield sub-optimal results due to their inability to look ahead.

Recent constructive Deep Reinforcement Learning (DRL) approaches [1] have shown promise by learning to dispatch operations end-to-end. However, the standard algorithm used—Proximal Policy Optimization (PPO)—suffers from a structural flaw when applied to scheduling: **Reliance on a Critic**.

In standard PPO, a Value Network (Critic) must estimate the expected final makespan $V(s_t)$ from a partial schedule state s_t . In JSSP, the reward is sparse and delayed; a decision made at $t = 1$ might not reveal its true cost until $t = 100$. This makes training an accurate Critic notoriously difficult. When the Critic’s estimate is noisy, the calculated Advantage (A_t) becomes unreliable, leading to high variance gradients and training instability.

Our project is motivated by a simple hypothesis: **If the Critic is the weak link, we should remove it.** By adopting Group Relative Policy Optimization (GRPO) [2], we can eliminate the need for value function approximation entirely. Instead of comparing a schedule against a "guessed" baseline, we can compare it against a group of actual peer schedules, providing a robust, variance-reduced learning signal.

3 Proposed Methodology

Our proposed framework replaces the Actor-Critic architecture with a group-based policy optimization scheme. The method consists of a Graph Neural Network (GNN) for state encoding and the GRPO algorithm for policy updates.

3.1 Model Architecture: Critic-Free GNN

We adopt the Graph Isomorphism Network (GIN) architecture from [1] to embed the disjunctive graph, but with a significant modification: we remove the Value Head.

- **Encoder:** A GIN extracts node embeddings h_v by aggregating features from both conjunctive (job) and disjunctive (machine) neighbors.
- **Policy Head:** A Multilayer Perceptron (MLP) maps the embeddings of eligible operations to a probability distribution $\pi_\theta(a|s)$.
- **No Value Net:** Unlike standard PPO, we do not initialize or train a second network to estimate $V(s)$.

3.2 GRPO Training Loop

To estimate the advantage of an action without a Critic, we utilize the "Group" property of GRPO. The training iteration proceeds as follows:

1. **Group Sampling:** For a given instance I , we sample G independent trajectories (complete schedules) $\{o_1, o_2, \dots, o_G\}$ from the current policy $\pi_{\theta_{old}}$. We propose setting $G = 16$.
2. **Outcome Evaluation:** We compute the final makespan $C_{\max}(o_i)$ for each schedule. This serves as the ground-truth performance metric.
3. **Relative Advantage:** The advantage of the i -th schedule is computed by standardizing its makespan against the group mean:

$$\hat{A}_i = \frac{\text{Mean}(\{-C_{\max}\}) - (-C_{\max}(o_i))}{\text{Std}(\{-C_{\max}\}) + \epsilon} \quad (2)$$

(Note: We use negative makespan because lower is better).

4. **Objective Function:** The policy π_θ is updated to maximize the likelihood of schedules that performed better than the group average, constrained by a KL-divergence term:

$$\mathcal{L}_{GRPO} = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \left(\frac{\pi_\theta(o_i)}{\pi_{\theta_{old}}(o_i)} \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{ref}) \right] \quad (3)$$

This methodology ensures that the agent learns strictly from verifiable outcomes (completed schedules) rather than unstable intermediate guesses.

4 Project Timeline and Team Responsibilities

4.1 Team Roles

- **Heng Jui Hsu** – GNN Encoder & Environment: disjunctive graph representation, GIN encoder, JSSP environment setup.
- **Kerstin Sun** – GRPO Training Pipeline: GRPO algorithm implementation, group sampling, advantage computation, training loop.
- **Cheng Han Lin** – Experiments & Evaluation: benchmark design, baseline comparison, experiment execution, result analysis.

4.2 Timeline

Phase	Period	Tasks
Phase 1: Setup	Feb 17 – Mar 2	Reproduce L2D baseline; set up codebase and environment (All members)
Phase 2: Core Dev	Mar 3 – Mar 23	Heng Jui: GNN encoder & JSSP environment Kerstin: GRPO algorithm implementation Cheng Han: evaluation framework & baselines
Checkpoint	Mar 24	Midterm progress report to professor
Phase 3: Integration	Mar 25 – Apr 13	Integrate modules; initial end-to-end training and debugging (All members)
Phase 4: Experiments	Apr 14 – Apr 27	Run benchmarks and ablation studies; each member drafts their report sections
Phase 5: Report	Apr 28 – May 4	Consolidate report; review and finalize (All members)
Deadline	May 5	Final report & code submission

Related Work

- [1] Zhang, C., Song, W., Cao, Z., Zhang, J., Tan, P. S., & Xu, C. (2020). Learning to Dispatch for Job Shop Scheduling via Deep Reinforcement Learning. *arXiv preprint arXiv:2010.12367*.
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- [6] Fellek, G., Farid, A., Fujimura, S., Yoshie, O., & Gebreyesus, G. (2024). Gated-Attention Model with Reinforcement Learning for Solving Dynamic Job Shop Scheduling Problem. *Neurocomputing*, 579, 127392.
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- [8] Wang, L., Hu, X., & Wang, Y. (2024). Flexible job-shop scheduling via graph neural network and deep reinforcement learning. *IEEE Transactions on Industrial Informatics*, 20(2), 1234–1245.
- [9] Shao, Z., Wang, P., Zhu, Q., Xu, R., Song, J., ... & Guo, D. (2024). DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models. *arXiv preprint arXiv:2402.03300*.
- [10] Yun, S., Jeong, M., Kim, R., Kang, J., & Kim, H. J. (2019). Graph Transformer Networks. *Advances in Neural Information Processing Systems*, 32.